

1. Objective

The aim of this lab was to apply text mining techniques to cleaned YouTube comments and captions so that the main themes, repeated expressions, and sentiment patterns could be compared in a structured way. Building on the earlier labs, this report focuses on TF-IDF keyword extraction, keyword overlap, bigram analysis, sentiment analysis, co-occurrence patterns, and a simple temporal split of the comment data. The main purpose was to understand how the spoken content of the video compares with what viewers focused on in the comment section.

2. Tools and Libraries Used

- Python 3.x: main programming language used for the implementation
- Pandas: loading CSV files, handling tables, and saving outputs
- Numpy: array splitting for temporal chunk analysis
- Matplotlib: generating keyword, bigram, sentiment, and temporal plots
- scikit-learn: TF-IDF vectorization and optional clustering support
- matplotlib-venn: creating the keyword overlap Venn diagram
- TextBlob: polarity-based sentiment analysis
- ast, os, re, collections.Counter, itertools.combinations: token parsing, file handling, counting, and co-occurrence analysis

3. Dataset Description

This lab used the cleaned datasets prepared from earlier work: `cleaned_comments.csv` and `cleaned_captions.csv`. The comments file contained 739 cleaned comment rows, while the captions file contained 187 cleaned caption rows. In both files, the `cleaned_tokens` column was used as the main input for text analysis. These tokens were joined back into normal text strings before TF-IDF and sentiment analysis were applied.

Table 1. Dataset summary used in Lab 6.

Dataset	Rows	Main Columns Used	Purpose in This Lab
<code>cleaned_comments.csv</code>	739	username, timestamp_text, comment_text, cleaned_tokens	Viewer discussion and audience reactions
<code>cleaned_captions.csv</code>	187	caption_sentence, cleaned_tokens	Spoken content from the video

4. Methodology / Procedure

4.1 Data Loading & Preparation

Both CSV files were loaded into pandas DataFrames. The script first checked whether the `cleaned_tokens` column existed. Because token columns in CSV files can sometimes be stored as plain strings, a safe parsing step was used to convert them into real token lists when needed. After validation, each token list was joined into a single `analysis_text` string so that TF-IDF vectorization and sentiment scoring could be run directly.

At the end of this step, two ready-to-use files were also saved in the output folder: `lab6_comments_ready.csv` and `lab6_captions_ready.csv`. These served as the prepared versions of the datasets for the remaining parts of the lab.

4.2 TF-IDF Keyword Extraction

TF-IDF was applied separately to comments and captions using `TfidfVectorizer` (`min_df=2`, `max_df=0.85`). The same settings were used for both datasets so that the results stayed comparable. After vectorization, the mean TF-IDF score for each term and its document frequency were calculated. The top 15 keywords were then saved to CSV files and plotted as horizontal bar charts.

This step was useful because it highlighted the words that were important within each dataset, not just the words that appeared many times. In other words, it reduced the influence of very common terms and brought forward the terms that were more informative for each collection of text.

4.3 Keyword & Theme Comparison

To compare the two datasets directly, the top 20 keywords from comments and the top 20 keywords from captions were collected and turned into sets. Their intersection and unique terms were then identified. A Venn diagram was created to show how many high-ranked keywords were shared and how many were unique to each side.

This comparison step helped answer a key question in the lab: whether the audience mainly repeated the same ideas expressed in the video, or whether the comments brought in additional themes not emphasized in the captions.

4.4 N-gram (Bigram) Analysis

After unigram-level TF-IDF, the same process was repeated using bigrams with `ngram_range=(2, 2)`. The top 10 bigrams were extracted separately for comments and captions. This was done because single-word keywords often show broad topics, while bigrams reveal more exact phrase-level patterns such as scientific terms, expressions, and repeated conceptual combinations.

The bigram outputs were saved both as CSV files and as bar charts so they could be compared in the report.

4.5 Sentiment Analysis

Sentiment analysis was performed with `TextBlob`. Each comment and each caption sentence was assigned a polarity score. The score was then mapped into one of three labels: positive, neutral, or negative. A threshold-based rule was used in the script so that small polarity values stayed in the neutral category rather than being forced into positive or negative unnecessarily.

After labeling, the sentiment counts and percentages were calculated for both datasets and plotted as bar charts. This allowed a simple comparison between creator-side language in the captions and viewer-side reactions in the comments.

4.6 Linking to Previous Labs

Lab 3 was linked through co-occurrence analysis. Using the top TF-IDF terms as the allowed term set, word pairs appearing together within the same token list were counted. The strongest co-occurring pairs were then saved and visualized for both comments and captions.

Lab 4 was linked through a temporal-style split of the comments dataset. The 739 comments were divided into five chunks using row order, and the frequencies of the top tracked keywords were observed across those chunks. This is only a simulated time view because the split was done by row position, not by exact chronological posting order.

Lab 5 was kept as an optional extension. The strict version of the script contains a clustering block, but it was left disabled during this run so that the report stayed focused on the required Lab 6 outputs.

5. Results and Discussion

5.1 TF-IDF Top Keywords

The TF-IDF results show that both datasets revolve around a very similar scientific topic. In both comments and captions, the highest-ranked word was universe, followed by infinite. This immediately suggests that the video topic strongly influenced the discussion. However, the rest of the rankings show an important difference between what was said in the video and what viewers chose to talk about afterward.

Table 2. Top 15 TF-IDF keywords in comments.

term	mean_tfidf	document_frequency
universe	0.059291	274
infinite	0.042508	156
reply	0.032848	133
like	0.030168	123
video	0.023232	66
time	0.022919	86
space	0.022559	78
big	0.020059	60
know	0.019442	54
one	0.018405	65
would	0.017710	65
edge	0.017241	56
think	0.015872	55
everything	0.015865	44
make	0.014958	50

As shown in Table 2 and Figure 1, the comments contained not only topic words such as universe, infinite, time, and space, but also audience-side words like reply, video, know, think, and everything. These terms suggest that viewers were not only discussing the scientific content itself; they were also reacting to the presentation, replying to one another, and expressing interpretation or opinion.

Table 3. Top 15 TF-IDF keywords in captions.

term	mean_tfidf	document_frequency
universe	0.068496	68
infinite	0.041460	23
could	0.038290	26
see	0.036188	26
edge	0.032816	14
time	0.030728	20
finite	0.029390	19
like	0.027717	20
would	0.022628	12
forever	0.022075	15
big	0.021364	16
mean	0.020191	14

observable	0.020023	12
way	0.019738	14
thing	0.019708	9

Table 3 and Figure 2 show a more focused vocabulary in the captions. In addition to universe and infinite, the captions emphasize could, see, edge, finite, forever, observable, and mean. This suggests that the video explanation was more descriptive and concept-driven, while the comments were slightly broader and more conversational.

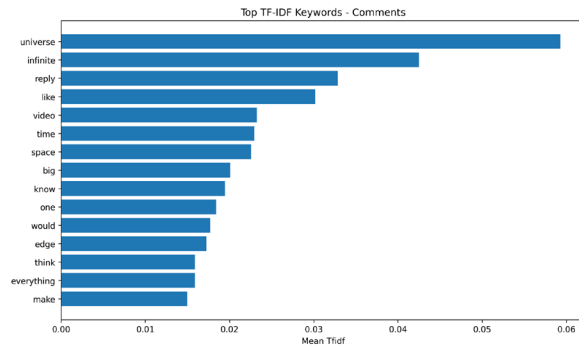


Figure 1. Top TF-IDF keywords in comments.

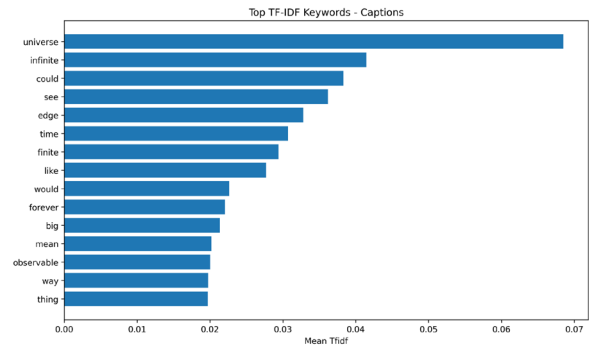


Figure 2. Top TF-IDF keywords in captions.

5.2 Keyword Overlap and Theme Comparison

Table 4. Top-20 keyword overlap summary.

Category	Count	Keywords
Common keywords	9	big, edge, finite, infinite, like, thing, time, universe, would
Comments only	11	brain, everything, infinity, know, make, never, one, reply, space, think, video
Captions only	11	also, bigger, could, find, forever, light, mean, observable, really, see, way

The keyword overlap comparison is summarized in Table 4 and visualized in Figure 3. Out of the top 20 keywords from each dataset, 9 were shared, while 11 were unique to comments and 11 were unique to captions. The shared set included big, edge, finite, infinite, like, thing, time, universe, and would.

This pattern shows that the two datasets are connected but not identical. The comments-only terms included brain, everything, infinity, know, make, never, one, reply, space, think, video. These look more audience-driven and reflective. In contrast, the captions-only terms included also, bigger, could, find, forever, light, mean, observable, really, see, and way, which sound closer to explanatory language used inside the video itself.

Top 20 Keyword Overlap: Comments vs Captions

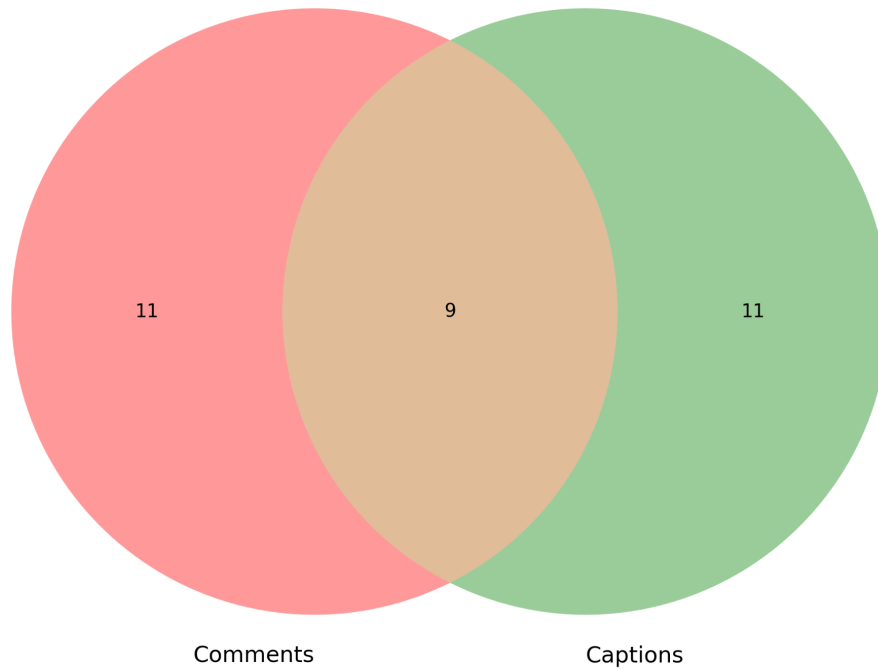


Figure 3. Venn diagram of top 20 keyword overlap between comments and captions.

5.3 Bigram Results

Table 5. Top 10 TF-IDF bigrams in comments.

term	mean_tfidf	document_frequency
big bang	0.021152	38
universe infinite	0.020781	40
infinite universe	0.017619	43
black hole	0.014638	21
observable universe	0.014454	23
edge universe	0.011392	17
dont know	0.009908	12
look like	0.008691	11
make sense	0.008348	15
feel like	0.008026	12

Table 6. Top 10 TF-IDF bigrams in captions.

term	mean_tfidf	document_frequency
make sense	0.019112	7
infinite universe	0.017360	8
universe infinite	0.016391	6
observable universe	0.015552	7
far away	0.015468	8
finite universe	0.012973	6
universe truly	0.011351	6
infinite space	0.011127	5
finite amount	0.011037	5
like universe	0.010016	3

The bigram results made the differences much clearer. In the comments, the strongest bigrams were big bang, universe infinite, infinite universe, black hole, and observable universe. These terms show that viewers were actively naming scientific ideas and connecting the video topic to familiar cosmology concepts.

In the captions, the top bigrams were make sense, infinite universe, universe infinite, observable universe, far away, finite universe, universe truly, and infinite space. Compared with the comments, the captions contain more explanatory phrases and more structured descriptive combinations.

There were only 4 shared bigrams between the top 10 lists: infinite universe, make sense, observable universe, universe infinite. This smaller overlap shows that phrase-level language differed more than single-word vocabulary.

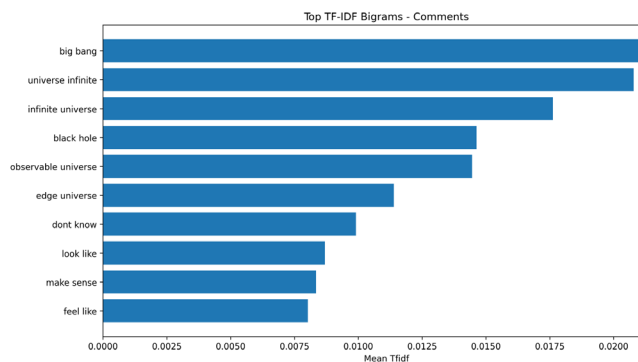


Figure 4. Top TF-IDF bigrams in comments.

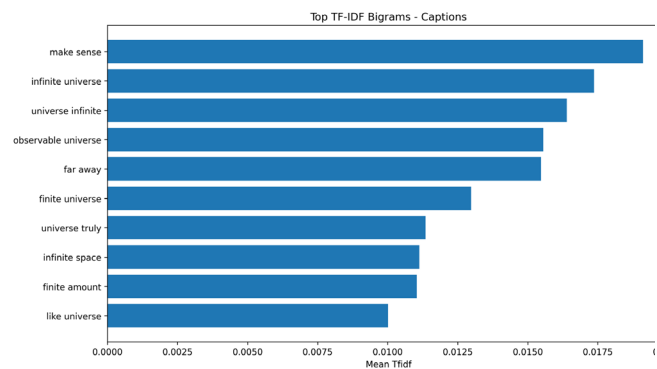


Figure 5. Top TF-IDF bigrams in captions.

5.4 Sentiment Results

Table 7. Sentiment distribution in comments and captions.

Sentiment	Comments Count	Comments %	Captions Count	Captions %
Positive	267	36.13	61	32.62
Neutral	384	51.96	104	55.61
Negative	88	11.91	22	11.76

The sentiment results are fairly balanced, but they still show a useful difference. As seen in Table 7, both datasets were mostly neutral. Comments were 51.96% neutral, 36.13% positive, and 11.91% negative. Captions were 55.61% neutral, 32.62% positive, and 11.76% negative.

This means the spoken content stayed slightly more neutral overall, which makes sense for an explanatory science video. The comments were still mostly neutral, but they showed a somewhat higher positive share. That suggests that audience responses included more admiration, agreement, or emotional engagement than the video narration itself.

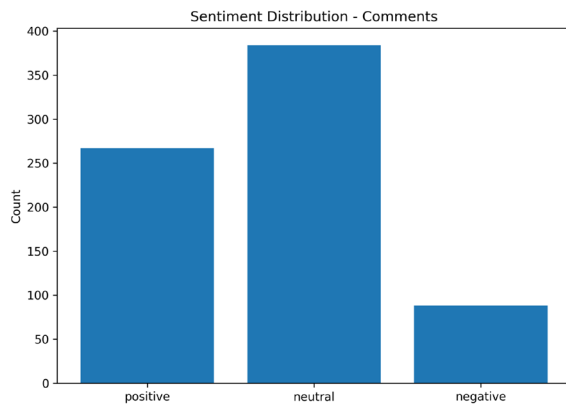


Figure 6. Sentiment distribution in comments.

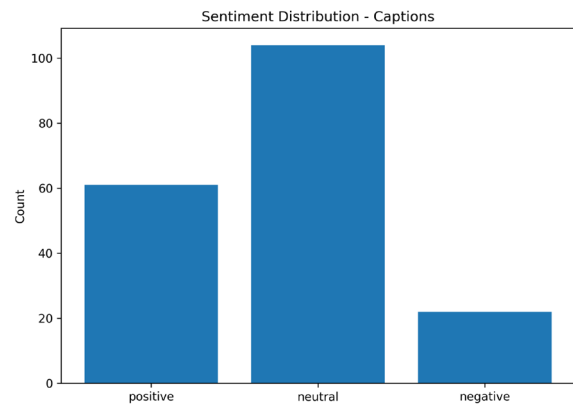


Figure 7. Sentiment distribution in captions.

5.5 Co-occurrence and Temporal Patterns

Table 8. Strongest co-occurring keyword pair in each dataset.

Dataset	Strongest Co-occurring Pair	Count
Comments	infinite + universe	112
Captions	infinite + universe	17

The co-occurrence analysis linked this lab directly to Lab 3. The strongest pair in both datasets was infinite + universe. However, the strength of that pair was much higher in comments (112) than in captions (17), which shows how strongly viewers kept repeating that central idea in different discussion lines.

Beyond the top pair, the comments frequently combined universe with like, time, reply, big, finite, and edge. In the captions, universe was commonly paired with could, see, observable, forever, big, finite, and time. This again supports the idea that captions stayed explanatory while comments became more conversational and interpretive.

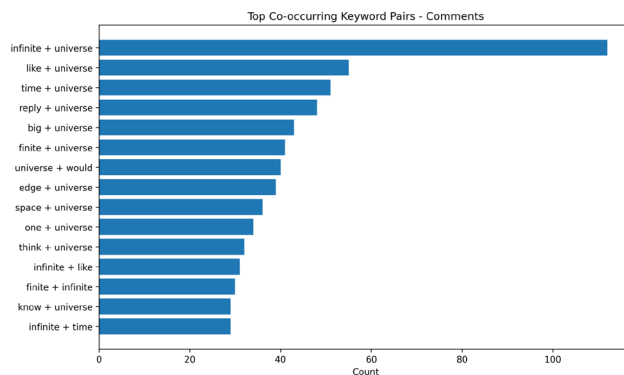


Figure 8. Top co-occurring keyword pairs in comments.

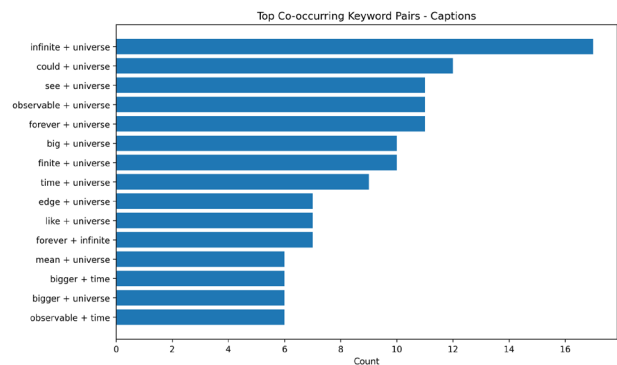


Figure 9. Top co-occurring keyword pairs in captions.

Table 9. Temporal-style keyword frequencies across five comment chunks.

chunk	rows_in_chunk	universe	infinite	reply	like	video
Chunk 1	148	49	34	63	19	12
Chunk 2	148	69	31	50	35	25
Chunk 3	148	91	53	20	35	23
Chunk 4	148	120	76	0	32	11
Chunk 5	147	110	54	1	28	10

The temporal-style split linked the work back to Lab 4. Table 9 and Figure 10 show that universe increased strongly from Chunk 1 to Chunk 4 before remaining high in Chunk 5. The word infinite also rose noticeably in the middle chunks and then fell back in the last chunk.

By contrast, reply started high and then dropped sharply, reaching zero in Chunk 4 and only 1 in Chunk 5. This suggests that some interaction-oriented discussion was concentrated earlier in the row order, while the core science theme around universe became even more dominant later in the dataset. The word like stayed fairly stable, and video remained a smaller but visible term throughout.

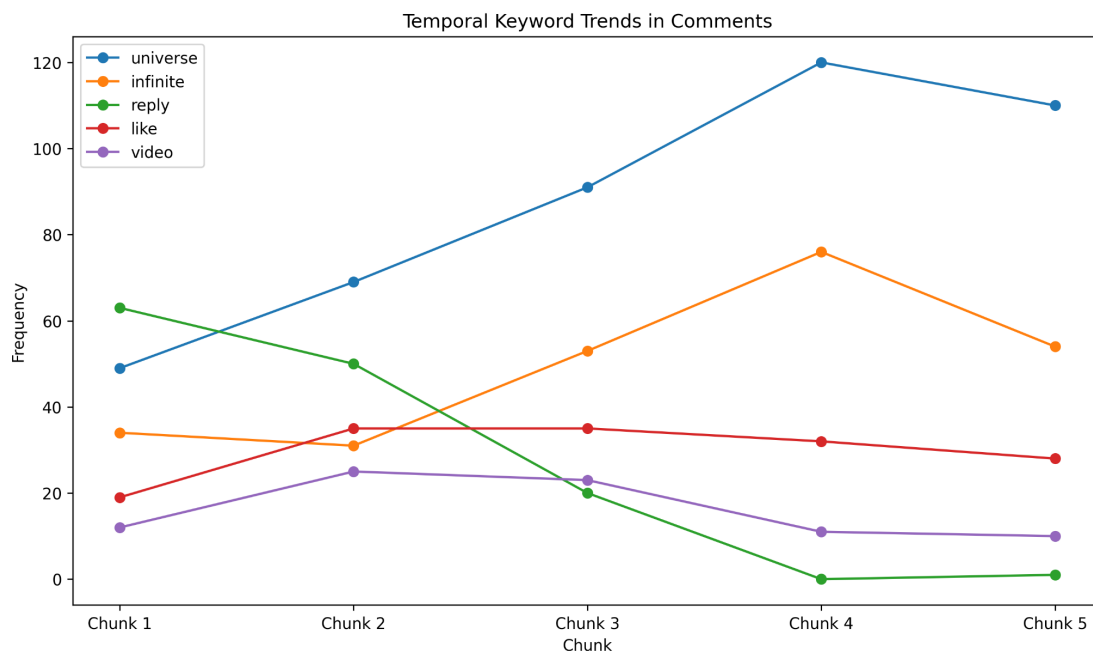


Figure 10. Temporal keyword trends in comments across five chunks.

6. Critical Prompts (Short Answers)

1. What themes did you expect to find before running TF-IDF?

Before running TF-IDF, I expected themes related to cosmology and the universe, especially words such as universe, space, time, infinite, edge, and big bang. Since the source video focused on scientific explanation, I also expected the comments to repeat some of those ideas while adding personal reactions.

2. Which dataset (captions or comments) showed more diverse language? Why?

The comments showed more diverse language. The captions were more focused and explanatory, while the comments mixed topic words with reaction words and interaction words such as reply, know, think, video, and everything. The keyword overlap results also showed that the comments introduced many terms that were not part of the main spoken vocabulary.

3. What do bigrams reveal that single keywords do not?

Bigrams reveal phrase-level meaning. A single word such as universe is broad, but a bigram such as big bang, black hole, make sense, or observable universe shows the exact concept or expression being used. This makes interpretation much clearer because it shows how the terms are actually connected in the text.

4. How do sentiment differences indicate audience reactions vs creator intentions?

The captions stayed slightly more neutral, which fits the creator's role of explaining a concept in a calm and informative way. The comments were still mostly neutral, but they had a somewhat higher positive share. That suggests the audience responded with more feeling, appreciation, curiosity, and personal interpretation than the original narration.

7. Conclusion

This lab successfully extended the earlier YouTube data work into a fuller text mining comparison between captions and comments. The results showed that both datasets shared a strong central focus on the universe and infinity, but they did not use language in exactly the same way. The captions were more focused on explanation and description, while the comments were broader, more conversational, and slightly more positive in sentiment.

The bigram, co-occurrence, and temporal results also added useful depth. They showed that phrase-level patterns and repeated word pairs gave a clearer picture than simple keyword lists alone, and the five-chunk comment split helped show how emphasis changed across the dataset.